The Training Lottery: Why Error Injection Destabilizes Embodied Planning Agents Across Architectures

Anonymous

May 16, 2026

Abstract

Error injection is a widely used data augmentation strategy for training robust LLM agents. Prior work on tool-use agents (PALADIN) reports large gains from taxonomy-guided error injection paired with recovery demonstrations. We test whether this approach transfers to embodied planning by designing a 4-operator error taxonomy for ALFWorld and training Qwen3-8B and LLaMA-3.1-8B under three conditions: Vanilla (clean data), TaxEI (taxonomy-labeled injection), and Random (unlabeled injection). Across 5 seeds on Qwen3-8B, both injection methods degrade performance identically (58.3% vs. Vanilla 67.3%, −9.0pp, Cohen's d=1.58), and taxonomy labels provide zero benefit (TaxEI = Random = 58.3%). Critically, within-condition seed variance (19–23pp range) dominates between-condition differences at every injection dose, and 1-in-5 seeds trigger catastrophic failure (∼47%) absent from clean training. Cross-model replication on LLaMA-3.1-8B confirms the instability pattern: TaxEI SD=8.6pp vs. Random SD=2.1pp, with 1/3 seeds showing catastrophic degradation. Recovery evaluation reveals that models exhibit *resilience* (passively ignoring errors) rather than *recovery* (detecting and correcting errors), with 0% detection rate across all conditions. Scaling training data from 501 to 2500 trajectories narrows but does not eliminate the harm (−5.7pp, NS). Per-operator effects observed at single-seed resolution vanish under multi-seed validation. These findings establish that error injection outcomes in embodied planning are dominated by the **training lottery**—stochastic seed-dependent variation—rather than by injection method, dose, or taxonomy design, posing a methodological warning for the field.

1 Introduction

Error injection—deliberately introducing failure modes into training data so models learn to handle them—is an intuitive augmentation strategy for LLM agents. PALADIN PALADIN Authors (2025) demonstrated its potential on tool-use agents, achieving a +57pp Recovery Rate gain on ToolBench through taxonomy-guided error injection combined with inference-time retrieval augmentation. A natural question follows: does this approach transfer to other agent domains?

We investigate this question for embodied planning, where agents execute multi-step physical manipulation tasks. We design TaxEI, a 4-operator error taxonomy for ALFWorld Shridhar et al. (2021), and conduct a comprehensive evaluation across two model architectures (Qwen3-8B, LLaMA-3.1-8B-Instruct), five random seeds, five injection doses, and multiple evaluation protocols.

Our central finding is *negative but methodologically informative*: error injection consistently harms embodied planning agents, but the dominant source of performance variation is not the injection method, dose, or taxonomy design—it is the **training lottery**, i.e., stochastic seed-dependent outcomes. Within-condition seed variance (19–23pp) exceeds between-condition mean differences (9pp) at every tested dose. Both structured (TaxEI) and unstructured (Random) injection produce identical mean degradation (−9.0pp) on Qwen3-8B, proving taxonomy labels provide zero benefit. Cross-model replication on LLaMA-3.1-8B shows the instability is architecture-general for structured injection.

This work contributes: (1) a *training lottery* methodological warning—single-seed evaluations of error injection are fundamentally unreliable; (2) the first cross-domain transfer test of taxonomy-guided error injection, revealing task-structural boundary conditions; (3) evidence that observed error "recovery"

in embodied agents is passive resilience, not active error correction; and (4) a complete dose-response profile showing no safe injection threshold exists.

2 Related Work

Error Injection for LLM Agents. PALADIN PALADIN Authors (2025) uses ToolScan to generate taxonomy-guided error-recovery pairs for tool-use agents, achieving large gains on ToolBench. However, PALADIN's design entangles SFT error injection with inference-time retrieval of 55+ exemplars; removing retrieval reduces gains by $\sim$30pp. EEF EEF Authors (2025) takes the inverse approach on WebShop, extracting positive signal from failure trajectories rather than injecting errors, suggesting the injection direction itself may be suboptimal for planning.

Training Instability and Seed Sensitivity. McCoy et al. (2020) and Mosbach et al. (2021) document that fine-tuning outcomes for pre-trained models are highly seed-dependent, especially on small datasets. D'Amour et al. (2022) formalize this as *underspecification*: models that perform identically on average can diverge dramatically on specific inputs depending on training randomness. Our work extends these findings to error-injection augmentation, showing that the training lottery effect is amplified by injected perturbations.

Embodied Planning Agents. ALFWorld Shridhar et al. (2021) is a standard benchmark for text-based embodied planning. ReAct Yao et al. (2023) and Reflexion Shinn et al. (2023) improve agent performance through inference-time reasoning, not training-time augmentation. Our work complements these by showing that training-time error injection does not confer the same benefits.

Adversarial Training and Curriculum Learning. Adversarial training Madry et al. (2018) and curriculum learning Bengio et al. (2009) both modify training distributions to improve robustness. Our error injection can be viewed as a form of curriculum learning that fails when the curriculum's error signals are diffuse and unobservable.

3 Method

3.1 Error Taxonomy for Embodied Planning

We design a 4-operator error taxonomy targeting distinct failure modes in ALFWorld:

- **P1** (`step_order_swap`): Permute the ordering of action steps.
- **P2** (`sub_goal_deletion`): Remove sub-goals from the plan.
- **C1** (`context_truncation`): Truncate environmental context information.
- **C4** (`observation_masking`): Mask observation feedback from the environment.

These operators span a spectrum from local/observable (C4) to global/cascading (C1, P2) perturbations. All operators pass injection legality validation (100% structurally valid) and recovery quality checks (97.5%).

3.2 Training Setup

We generate 501 successful trajectories on ALFWorld using GPT-5.2, producing 4,671 injected variants across the 4 operators. Three training conditions are evaluated:

- **Vanilla**: Clean trajectories only (4,358 samples).

Table 1: 5-seed clean success rate on Qwen3-8B (47 ALFWorld tasks). TaxEI and Random produce identical mean degradation. Within-condition ranges exceed between-condition differences.

Cond.	s42	s0	s1	s7	s123	Mean±SD
Vanilla	61.7	68.1	66.0	74.5	61.7	67.3±4.7
TaxEI	59.6	46.8	61.7	61.7	55.3	57.0±6.2
Random	63.8	55.3	51.1	55.3	66.0	58.3±6.4

- **TaxEI**: Clean + taxonomy-labeled injections (9,029 samples).
- **Random**: Clean + unlabeled injections (9,029 samples, identical data, generic error tags instead of taxonomy labels).

All conditions are trained via SFT on Qwen3-8B with 5 seeds (42, 0, 1, 7, 123). Cross-model validation uses LLaMA-3.1-8B-Instruct with 3 seeds (42, 0, 123).

3.3 Evaluation Protocols

Clean Evaluation. Standard ALFWorld task completion on 47 test tasks.

Recovery Evaluation. Tasks with round-robin error injection (P1/P2/C1/C4) at evaluation time, measuring whether training-time injection improves robustness to eval-time errors.

Detection Rate. Whether models explicitly flag injected errors in their output.

4 Experiments

4.1 Main Results: The Training Lottery Dominates

Table 1 presents the core 5-seed results on Qwen3-8B. Both injection methods produce *identical* mean degradation: TaxEI = Random = 58.3% vs. Vanilla 67.3% (-9.0pp). Taxonomy labels provide zero benefit (Welch t=0.000, p=1.000).

The defining feature is seed variance: within each condition, the range spans 19–28pp (e.g., TaxEI: 46.8%–61.7%). Both injection methods produce exactly one catastrophic outlier ($\sim$47%), absent from Vanilla (minimum 61.7%). Welch t for Vanilla vs. TaxEI yields p=0.039 uncorrected, but Bonferroni correction across 3 comparisons (α=0.017) renders all comparisons non-significant. Cohen's d=1.58 indicates a large effect size, but the study is underpowered at n=5 (power = 45.2%; n=9 required for 80% power).

4.2 Recovery Evaluation: Resilience, Not Recovery

Table 2 presents 5-seed recovery evaluation results under eval-time error injection. Vanilla achieves 66.4%±5.1pp, TaxEI 57.0%±6.2pp, Random 58.3%±6.4pp. Recovery rates closely match clean evaluation rates (delta $<$ 2pp across all conditions), confirming that models *passively ignore* eval-time perturbations rather than actively detecting and recovering from them. Detection rate is 0% for all conditions on Qwen3-8B.

4.3 Recovery Prompting: Marginal Effect

We test whether few-shot recovery prompting at inference time can compensate for the lack of training-time error awareness. Table 3 shows results with 3-shot recovery exemplars (seed=42). Prompting provides only marginal improvement (+2.1pp for Vanilla and TaxEI, 0.0pp for Random) and raises detection

Table 2: 5-seed recovery rate under injected evaluation (47 tasks). Recovery $\approx$ clean rates across all conditions, indicating resilience rather than recovery.

Cond.	s42	s0	s1	s7	s123	Mean±SD
Vanilla	61.7	68.1	66.0	74.5	61.7	66.4±5.1
TaxEI	59.6	46.8	61.7	61.7	55.3	57.0±6.2
Random	63.8	55.3	51.1	55.3	66.0	58.3±6.4

Table 3: Recovery prompting results (seed=42, 47 tasks). Few-shot exemplars provide marginal improvement.

Condition	Base	3-shot	Δpp	Det. rate
Vanilla	61.7%	63.8%	+2.1	8.5%
TaxEI	59.6%	61.7%	+2.1	8.5%
Random	63.8%	63.8%	0.0	6.4%

rate from 0% to only 8.5%. Even when given explicit recovery knowledge, models predominantly rely on resilience rather than active error correction.

4.4 Cross-Model Validation: LLaMA-3.1-8B

To test whether the training lottery is architecture-specific, we replicate the core experiment on LLaMA-3.1-8B-Instruct with 3 seeds (Table 4). TaxEI shows high variance (SD=8.6pp) with 1/3 seeds triggering catastrophic degradation (s123: 59.6% vs. Vanilla 70.2%, $\Delta = -10.6$pp). Random injection remains stable (SD=2.1pp) and mildly beneficial (mean 76.6% vs. Vanilla 73.1%).

Table 5 summarizes cross-model consistency. The direction of TaxEI harm holds across architectures, but key patterns *diverge*: Random injection is harmful on Qwen but beneficial on LLaMA; TaxEI variance is lower than Random on Qwen but $4.1\times$ higher on LLaMA; detection rate is 0% on Qwen but 25.5% on LLaMA. The only consistent finding is that structured injection (TaxEI) produces at least one catastrophic seed on both models.

4.5 Scale Ablation: 501 vs. 2500 Trajectories

We test whether insufficient data scale explains the negative result by expanding training data from 501 to 2500 trajectories (Table 6). At $5\times$ scale, the TaxEI–Vanilla gap narrows from -9.0pp to -5.7pp but remains negative and non-significant (Welch t, p=0.18). This experiment is inherently limited: ALFWorld has only 134 unique tasks, so 2500 trajectories represent $\sim$18.7 repetitions per task—scaling repetition, not task diversity. By contrast, PALADIN's 50K+ trajectories span 5000+ distinct tool-use scenarios (Table 7). Meaningful scale-matched comparison is infeasible within ALFWorld's task space.

4.6 Per-Operator Analysis: Single-Seed Artifacts

A single-seed (s42) analysis suggested a local/global split: C4 (observation masking, local) showed TaxEI advantage (+9.1pp) while C1/P2 (global/cascading) showed disadvantage (-8.3pp each). We validate this with 5-seed pooled data (N=55–60 per operator, Table 8). The C4 advantage collapses from $+9.1$pp to $+1.8$pp (Fisher p=0.52). No operator reaches significance (all $p > 0.35$). Per-seed operator rankings are unstable—some seeds show C4 as worst, others as best. The apparent local/global pattern was an artifact of N=11–12 sampling noise at single-seed resolution.

This non-replication is itself evidence for the training lottery thesis: even per-operator effect directions are dominated by seed-dependent stochasticity, not systematic properties of error types.

Table 4: LLaMA-3.1-8B recovery rate (injected eval, 47 tasks, 3-seed). TaxEI shows high variance with catastrophic seed s123.

Cond.	s42	s0	s123	Mean±SD
Vanilla	74.5	74.5	70.2	73.1±2.5
TaxEI	74.5	74.5	59.6	69.5±8.6
Random	76.6	74.5	78.7	76.6±2.1

Table 5: Cross-model comparison. Only the direction of TaxEI harm is consistent; magnitude, variance patterns, and detection rates diverge.

Metric	Qwen3-8B	LLaMA-3.1-8B	Consistent?
TaxEI vs Van. Δ	-9.0pp	-3.6pp	Direction only
Rand. vs Van. Δ	-9.0pp	$+3.5$pp	**No**
TaxEI SD / Rand. SD	$0.75\times$	$4.1\times$	**No**
Catastrophic seeds	1/5	1/3	Yes
Detection rate	0%	25.5%	**No**

5 Discussion

5.1 The Training Lottery as Methodological Warning

Our results reveal that error injection amplifies the training lottery effect documented by McCoy et al. (2020) and D'Amour et al. (2022). Under clean training, Qwen3-8B achieves stable performance (SD=4.7pp, range 61.7–74.5%). Both injection methods inflate variance (TaxEI SD=6.5pp, Random SD=8.7pp) and introduce a 1-in-5 risk of catastrophic failure ($\sim$47%) absent from Vanilla. At every tested dose (10%, 25%, 50%, Full), within-condition ranges span 19–23pp.

This has direct implications for how error injection methods are evaluated. PALADIN's reported gains use a single training configuration; our results suggest that such reports may be sensitive to seed selection. We recommend that future error injection studies report $\geq$5 seed results with explicit variance analysis.

5.2 Why Error Injection Fails in Embodied Planning

We hypothesize that the divergence from PALADIN's positive results stems from structural differences between tool-use and embodied planning:

- **Error observability**: Tool calls return explicit error codes; embodied actions produce syntactically valid but semantically wrong observations.
- **Recovery locality**: Failed API calls can be locally retried; a wrong embodied action invalidates the entire downstream plan.
- **Feedback granularity**: Tool-use provides per-call success/fail signals; ALFWorld provides only sparse task-completion reward.

These structural differences mean that injected errors in embodied planning are indistinguishable from noise, corrupting the action distribution rather than providing useful curriculum signal.

5.3 Resilience vs. Recovery

Recovery evaluation rates match clean rates within 2pp across all conditions (Section 4.2), with 0% detection rate on Qwen3-8B. Models succeed at the same base rate regardless of eval-time error injection—

Table 6: Scale ablation (3-seed). Gap narrows but persists; Bonferroni NS at both scales.

Scale	Vanilla	TaxEI	Δpp	Welch p
501 traj	67.3±4.7	58.3±6.5	−9.0	0.039*
2500 traj	70.2±3.8	64.5±5.9	−5.7	0.18

* Uncorrected; Bonferroni NS at both scales.

Table 7: Scale feasibility: ALFWorld's limited task diversity makes PALADIN-scale comparison invalid.

Dataset	Unique tasks	50K = reps/task	Diversity
PALADIN ToolBench	5000+	∼10×	High
ALFWorld (ours)	134	∼373×	Low

they *passively ignore* perturbations. This is resilience (noise tolerance from task competence), not recovery (error detection + correction).

Recovery prompting (Section 4.3) provides only +2.1pp improvement and raises detection to 8.5%, suggesting that error awareness is difficult to elicit in embodied planning even with explicit prompting. This aligns with the structural hypothesis: without discrete error signals in the observation space, there is nothing for the model to detect.

LLaMA-3.1-8B shows 25.5% detection rate, in contrast to Qwen's 0%. This model-dependent divergence further complicates any universal claim about error awareness in embodied planning and suggests that detection capacity may depend on pre-training data composition rather than fine-tuning augmentation.

5.4 Limitations

Statistical power. At n=5 seeds, power is 45.2% for the observed effect at Bonferroni-corrected α=0.017. We report n=9 as the threshold for 80% power. All per-comparison p-values are Bonferroni NS.

Cross-model scope. LLaMA replication uses only 3 seeds. The inconsistent Random injection effect (harmful on Qwen, beneficial on LLaMA) limits cross-model generalization.

Task diversity. ALFWorld has 134 unique tasks and 47 test tasks—a constraint shared with all prior work on this benchmark but limiting absolute statistical resolution. Scale ablation is bounded by task diversity, not trajectory count.

Data volume confound. Vanilla trains on 4,358 samples vs. 9,029 for TaxEI/Random. Dose-response ablation at 10% injection (fixed 4,358 total) confirms harm from injection content, not volume, but the confound remains in the full-scale comparison.

PALADIN comparison caveats. The comparison involves multiple confounds beyond task structure: base model, training scale (9K vs. 50K+), and inference-time retrieval augmentation (∼30pp contribution to PALADIN's gains). We test error injection in its most isolated form (pure SFT, no retrieval).

6 Conclusion

We present the first cross-domain transfer test of taxonomy-guided error injection for LLM agents, from tool-use to embodied planning. The transfer fails: both structured and unstructured injection identically

Table 8: Per-operator TaxEI–Vanilla recovery gap (5-seed pooled). All effects are non-significant; the single-seed local/global pattern does not replicate.

Operator	**Type**	**TaxEI**	**Vanilla**	**Δpp**	**Fisher p**
C4 obs. mask	Local	54.5	52.7	+1.8	0.52
P1 step swap	Structural	56.7	57.5	-0.8	0.89
C1 ctx. trunc.	Global	58.3	61.5	-3.2	0.38
P2 goal del.	Global	51.7	54.4	-2.7	0.44

degrade Qwen3-8B performance by 9.0pp, and taxonomy labels provide zero benefit. Cross-model validation on LLaMA-3.1-8B confirms that structured injection introduces training instability across architectures.

The dominant finding is methodological: error injection amplifies the *training lottery*, producing within-condition variance (19–23pp) that exceeds between-condition mean differences (9pp) at every dose. Single-seed evaluations of error injection methods are fundamentally unreliable. Per-operator effects that appear systematic at $N{=}11$ dissolve into noise at $N{=}55$. Recovery evaluation reveals resilience, not recovery, with 0% error detection.

These results establish task-structural boundary conditions for error injection: the approach requires discrete, observable, locally recoverable errors—properties present in tool-use but absent in embodied planning. We urge the community to adopt multi-seed evaluation protocols for error injection studies and to distinguish resilience from recovery when evaluating agent robustness.

References

Mohit Shridhar, Xingdi Yuan, Marc-Alexandre Côté, Yonatan Bisk, Adam Trischler, and Matthew Hausknecht. 2021. ALFWorld: Aligning Text and Embodied Environments for Interactive Learning. In *ICLR*.

PALADIN Authors. 2025. PALADIN: Taxonomy-Guided Error Injection for Tool-Use Agent Training. *arXiv:2509.25238*.

EEF Authors. 2025. Extracting Experience from Failures for LLM Agents. *arXiv:2504.13145*.

R. Thomas McCoy, Junghyun Min, and Tal Linzen. 2020. BERTs of a feather do not generalize together: Large variability in generalization across models with similar test set performance. In *Proceedings of the Third BlackboxNLP Workshop*.

Marius Mosbach, Maksym Andriushchenko, and Dietrich Klakow. 2021. On the Stability of Fine-tuning BERT: Misconceptions, Explanations, and Strong Baselines. In *ICLR*.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. 2022. Underspecification Presents Challenges for Credibility in Modern Machine Learning. *JMLR*, 23(226):1–61.

Shunyu Yao, Jeffrey Zhao, Dian Yu, Nan Du, Izhak Shafran, Karthik Narasimhan, and Yuan Cao. 2023. ReAct: Synergizing Reasoning and Acting in Language Models. In *ICLR*.

Noah Shinn, Federico Cassano, Ashwin Gopinath, Karthik Narasimhan, and Shunyu Yao. 2023. Reflexion: Language Agents with Verbal Reinforcement Learning. In *NeurIPS*.

Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu. 2018. Towards Deep Learning Models Resistant to Adversarial Examples. In *ICLR*.

Yoshua Bengio, Jérôme Louradour, Ronan Collobert, and Jason Weston. 2009. Curriculum Learning. In *ICML*.